

EXPERIENCE

Rural Studies Institute, GCSU (Sponsored by Georgia Tech PIN Program)

Milledgeville, GA

Full-Stack Developer & Data Pipeline Intern

May 2025 – Aug 2025

- Developed custom **Python** scrapers to aggregate socio-economic datasets from US Census (ACS) and USDA APIs, automating ingestion of **10k+** data points and cutting manual data entry by **40%**.
- Built automated data pipelines converting raw spreadsheets to **GeoJSON**, reducing integration time by **30%**.
- Designed and deployed interactive map-based web platform using JavaScript, Leaflet.js, and WordPress; launched to **500+** monthly users across research and policy teams.
- Collaborated with researchers and policymakers to translate technical implementations into accessible, production-ready tools.
- Presented at the *Emerging Innovators Celebration*.
- **Project Link:** gcsu.edu/ruralstudies/understanding-black-belt-region

TECHNICAL SKILLS

- **Core (Proficient):** JavaScript/TypeScript, Python, Node.js, Express.js, React.js, MongoDB, Git, REST APIs
- **Backend & Cloud:** AWS (EC2, S3, Lambda, RDS, IAM), Docker, Linux, CI/CD (GitHub Actions), WebSockets/Socket.io, JWT, Vercel, Render
- **Databases & Data:** MongoDB, PostgreSQL, Prisma, SQL, schema design, indexing
- **Testing & Quality:** Jest, Vitest, Unit & Integration Testing
- **Frontend:** Next.js, Tailwind CSS, Recoil, Zustand, Vite, HTML/CSS
- **Tools & Practices:** Postman, Zod, Bcrypt, Agile/Scrum, Code Reviews, Technical Documentation

PROJECTS

Real-Time Collaborative Task Board | *React, Node.js, Express, Socket.io, MongoDB, JWT* [GitHub](#) | [Demo](#)

- Built real-time collaboration system using **WebSockets**, enabling instant task updates across **50+** concurrent users with **<100ms** latency.
- Designed **RESTful API** with **JWT** authentication, **Zod** validation, **Bcrypt** password hashing, and role-based access control.
- Improved data consistency by implementing a conflict resolution strategy for concurrent edits, reducing state synchronization errors by **15%** during peak activity.
- Deployed on **Vercel** (frontend) and **Render** (backend) with automated **CI/CD via GitHub Actions** and health monitoring.

TweenGen | *React.js, Fabric.js, Recoil, Vite*

[GitHub](#) | [Demo](#)

- Architected a lightweight animation engine with optimized easing algorithms, reducing the final bundle size by **15%** compared to standard external animation libraries.
- Reduced canvas re-render cycles by **40%** through memoization and selective state updates in Recoil, maintaining a consistent **60 FPS** during complex multi-object animations.
- Built drag-and-drop timeline UI with GSAP-powered live preview and code export system generating production-ready **HTML/CSS/JavaScript**. Hosted on **Vercel**.

EDUCATION

The University of Georgia

Athens, GA

Master of Science in Computer Science

Aug 2024 – May 2026

- Coursework: Software Engineering, Database Management, Algorithms, Computer Networks, Data Science, Advanced Representation Learning

Sathyabama Institute of Science and Technology

Chennai, India

Bachelor of Engineering in Computer Science

Aug 2020 – May 2024

CERTIFICATIONS

- AWS Certified Developer – Associate ([link](#))
- AWS Certified Cloud Practitioner ([link](#))